

HATS-2b

R-2265

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATS-2_220318 · created 2026-08-16T01:45:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459656.7916 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.031
Transit depth (Rp/Rs)^2	1e-06 +/- 0.00064 [%]
Orbital Inclination (inc)	84.42 +/- 2.11
Airmass coefficient 1 (a1)	0.382 +/- 0.017
Airmass coefficient 2 (a2)	0.543 +/- 0.093
Scatter in the residuals of the lightcurve fit is	4.49 %
Best Comparison Star	None
Optimal Aperture	65.98
Optimal Annulus	171.72
Transit Duration (day)	0.0656 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.031 vs 0.1335 (deviation 4.31 σ)
Duration fit vs published	0.0656 d vs 0.0871 d (deviation 2.26 σ)
Mid-time O–C	5.4 min
Depth SNR	0.0
Residual scatter	4.49 %

Light curve

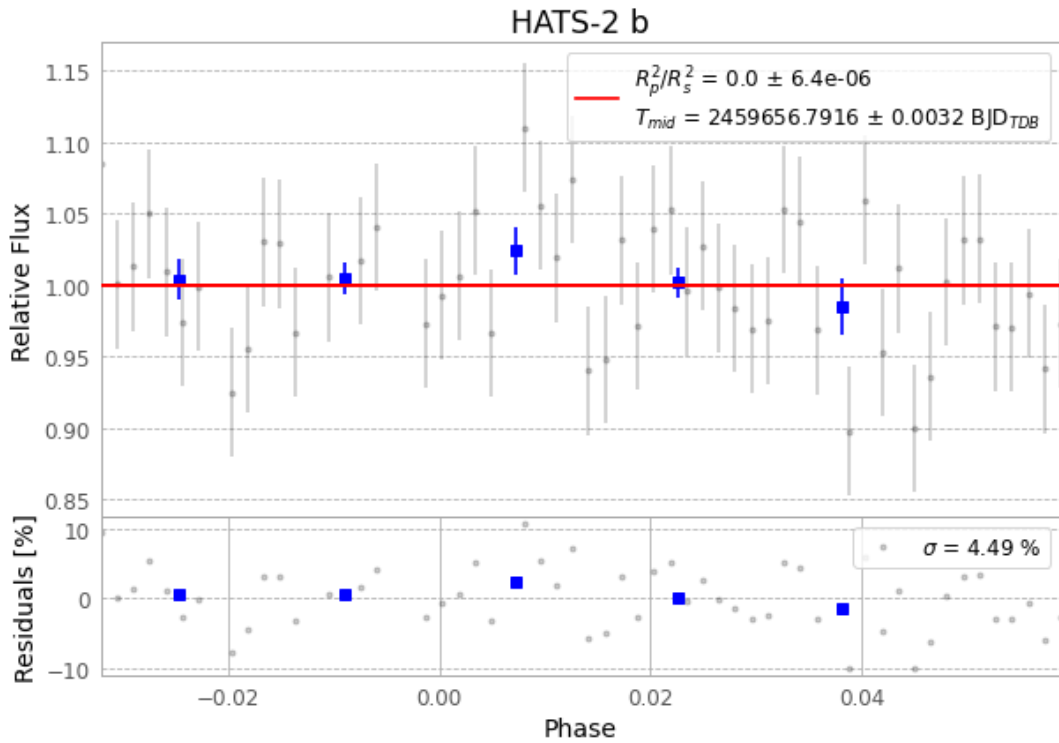


Figure 1 — FinalLightCurve_HATS-2 b_2022-03-17.png (SHA-256 9e38c212f90962a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 98], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02ecf5fc5cad190f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

60 FITS frames (39 MB), HATS-2220318054813.FITS → HATS-2220318084513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATS-2-220318.log (bf6f1a9b3ba9...), FinalLightCurve_HATS-2 b_2022-03-17.png (9e38c212f909...), FinalLightCurve_HATS-2 b_2022-03-17.csv (14f7a09d2351...)

Compute resources

Wall time 441.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:45Z.